

## Oscillations and Waves.

**Periodic motion:** It is the motion which gets repeated after some fixed interval of time. The time after which the motion is repeated is known as time period.

e.g.; - Motion of earth around the sun is a periodic motion with a period of 365 days.

- Rotation of earth about its own axis is a periodic motion with a period of 24 hours.
- Motion of moon around the earth is a periodic motion with a period of 27.3 days.
- Human heartbeat is periodic with a period of 0.8 second (Normally).

**NOTE:** The reciprocal of the time period is known as frequency.

e.g.; frequency of heartbeat  $\nu = \frac{1}{T} = \frac{1}{0.8} = \frac{10}{8}$

$= 1.25 \text{ Hz}$

**Periodic function:** is a function whose value gets repeated after some fixed interval and the fixed interval is known as period.

Examples: 1  $\sin \theta$ .

$$\sin(\theta + 2\pi) = \sin \theta$$

$$\rightarrow \sin(30^\circ + 2\pi) = \sin 30^\circ = \frac{1}{2}$$

$$\rightarrow \sin(45^\circ + 2\pi) = \sin 45^\circ = \frac{1}{\sqrt{2}}$$

Clearly  $\sin \theta$  is a periodic function with a period of  $2\pi$  i.e.; value of  $\sin \theta$  gets repeated after a period of  $2\pi$ .



2.  $\cos \theta$  and  $\cos(\theta + 2\pi)$  have same value.

-  $\theta = 30^\circ$

$\therefore \cos 30^\circ = \cos(30^\circ + 2\pi)$

$\frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}$

clearly,  $\cos \theta$  is a periodic function with a period of  $2\pi$ .

NOTE:  $\sin \theta$  is also a periodic function with a period of  $\pi$  radian ( $180^\circ$ ).

### oscillatory motion.

It is the motion in which a particle moves to and fro or back and forth about a fixed point known as mean position. The two end points between which its motion remains confined are known as extreme positions.

NOTE: All the oscillatory motions are periodic but all periodic motions are not oscillatory.

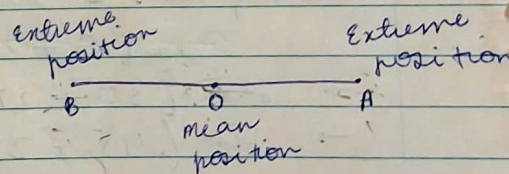
### one oscillation

O to A

A to O

O to B

B to O





Time period: time taken to complete one oscillation.

At  $t = 0$ , particle is at mean position O.

At  $t = \frac{T}{4}$ , particle is at extreme position A.

At  $t = \frac{T}{2}$ , particle is at mean position O.

At  $t = \frac{3T}{4}$ , particle is at extreme position B.

At  $t = T$ , particle is at mean position O.

Linear frequency  

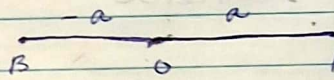
$$\nu = \frac{1}{T}$$

(a)

Angular frequency  

$$\omega = \frac{2\pi}{T} = 2\pi\nu$$

Amplitude: It is defined as the maximum displacement of the particle from the mean position. clearly, it is the distance between the mean position and the extreme ~~end~~ position.

Phase ( $\theta$ ): It gives us the info.  about ~~phase~~, position of the particle.

Angle assigned to one oscillation is  $2\pi$  radians.

At  $\theta = 0 \rightarrow$  particle at mean position O.

At  $\theta = \frac{\pi}{2} \rightarrow$  " " extreme position A.

At  $\theta = \pi \rightarrow$  " " mean position O.

At  $\theta = \frac{3\pi}{2} \rightarrow$  " " extreme position B.

At  $\theta = 2\pi \rightarrow$  " " mean position O.

Restoring force: In oscillatory motion, there is a force acting on the particle which is directed towards the mean position. This force is known as restoring force. It is due to this force that



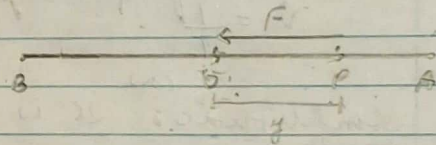
particle returns back from extreme position towards mean position.

Simple harmonic motion (SHM).

It is an oscillatory motion in which the restoring force acting on the particle is directly proportional to its displacement from the mean position.

$$F \propto y$$

$$F = -ky$$

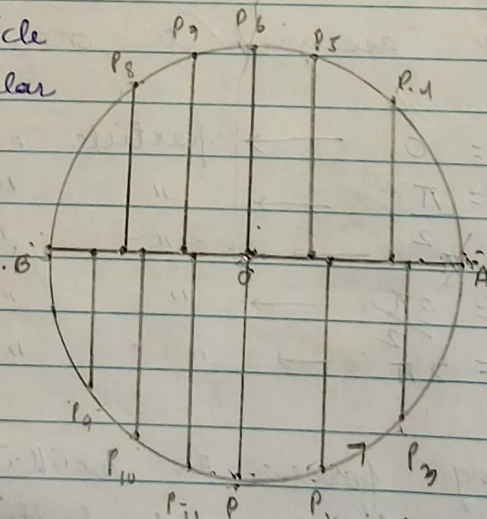


$$F \propto y \quad F = -ky$$

Here, negative sign indicates that the restoring force is directed opposite to displacement.

1 equation for displacement.

when a particle is in uniform circular motion completes one circle, its projection point completes one oscillation along one oscillation.



uniform circular motion



we have,

$$\omega = \frac{\theta}{t}$$

$$\theta = \omega t$$

Now, In  $\triangle OPP_2$ ,

$$\sin \theta = \frac{OP}{OP_2}$$

$$\sin \theta = \frac{y}{a}$$

$$y = a \sin \theta$$

But  $\theta = \omega t \rightarrow$  known as phase

$$\therefore y = a \sin \omega t$$

$$y = a \sin \omega t$$

At  $t = 0$ ,

$$y = a \sin 2\pi \times 0$$

$$y = a \sin 0 = a(0)$$

$$\boxed{y = 0}$$

At  $t = \frac{T}{4}$ ,

$$y = a \sin \frac{2\pi}{T} \times \frac{T}{4} = a \sin \frac{\pi}{2} = a \sin(90^\circ)$$

$$y = a(1) = a$$

$$\boxed{y = a}$$

At  $t = \frac{T}{2}$ ,

$$y = a \sin \frac{2\pi}{T} \times \frac{T}{2} = a \sin \pi = a(0)$$

$$\boxed{y = 0}$$



$$\text{At } t = \frac{3T}{4}$$

$$y = a \sin \frac{2\pi}{T} \times \frac{3T}{4}$$

$$y = a \sin \frac{3\pi}{2}$$

$$y = a (-1)^2$$

$$\boxed{y = -a}$$

$$\text{At } t = T$$

$$y = a \sin \frac{2\pi}{T} \times T$$

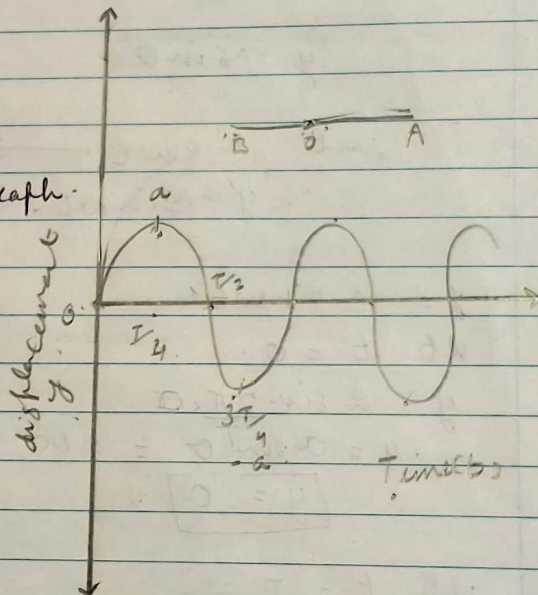
$$y = a \sin 2\pi$$

$$y = a (0)$$

$$\boxed{y = 0}$$

Displacement-time graph.  
(y vs t).

Graph is sinusoidal.



Resemblance

At  $\theta =$  value of  $\sin \theta$

$$\theta = 0 \quad \sin 0 = 0$$

$$\theta = \frac{\pi}{2} \quad \sin \frac{\pi}{2} = 1$$

$$\theta = \pi \quad \sin \pi = 0$$

$$\theta = \frac{3\pi}{2} \quad \sin \frac{3\pi}{2} = -1$$

$$\theta = 2\pi \quad \sin 2\pi = 0$$



Equation for velocity.

we have,

$$y = a \sin \omega t$$

Now,

$$v = \frac{dy}{dt}$$

$$v = \frac{d(a \sin \omega t)}{dt}$$

$$v = a \frac{d(\sin \omega t)}{dt}$$

$$v = a \cos \omega t \cdot \omega$$

$$v = a \omega \cos \omega t$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$v = \omega a \sqrt{\cos^2 \omega t}$$

$$v = \omega a \sqrt{1 - \sin^2 \omega t}$$

$$v = \omega \sqrt{a^2 - a^2 \sin^2 \omega t}$$

$$v = \omega \sqrt{a^2 - y^2}$$

At extreme positions

$$y = \pm a$$

$$v = \omega \sqrt{a^2 - a^2}$$

$$v = 0$$

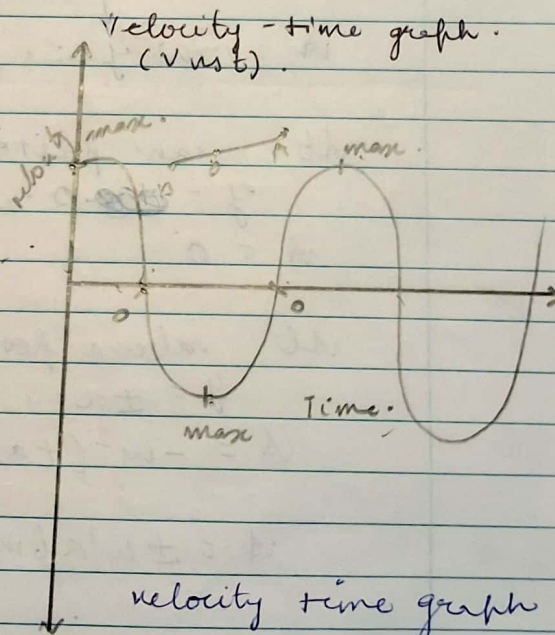
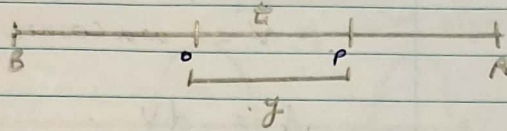
At mean position

$$y = 0$$

$$v = \omega \sqrt{a^2 - 0}$$

$$v = \omega a$$

$$v = \omega a \text{ --- maximum velocity}$$



velocity time graph is sinusoidal.



equation for acceleration (A).

we have,

$$v = wa \cos \omega t$$

$$A = \frac{dv}{dt}$$

$$A = \frac{d}{dt} (wa \cos \omega t)$$

$$A = wa \frac{d}{dt} (\cos \omega t)$$

$$A = wa (-\sin \omega t) \omega$$

$$A = -\omega^2 a \sin \omega t$$

$$A = -\omega^2 y \rightarrow \text{acceleration and displacement are opposite.}$$

At mean positions.

$$y = 0$$

$$A = 0$$

At extreme positions

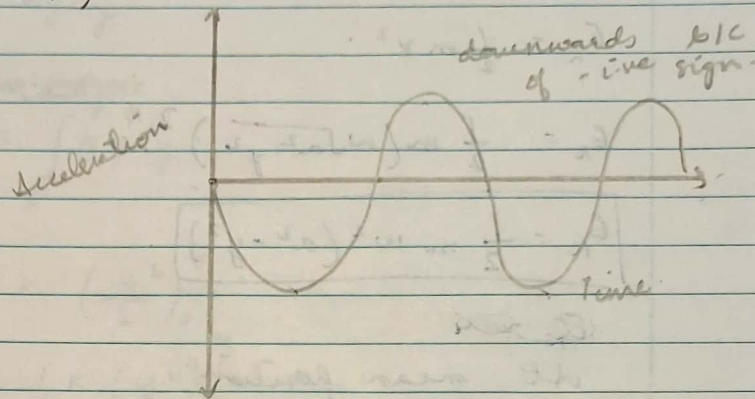
$$y = \pm a$$

$$A = -\omega^2 (\pm a)$$

$$A = \pm \omega^2 a \text{ (maximum).}$$



Acceleration - time graph.  
(A vs T).



Equation for time period. (T).

we have.

$$F = -ky.$$

But  $F = mA$

$$F = m(-\omega^2 y).$$

$$F = -m\omega^2 y.$$

$$-ky = -m\omega^2 y$$

$$\boxed{k = m\omega^2}$$

$$\omega^2 = \frac{k}{m}$$

$$\omega = \sqrt{\frac{k}{m}}$$

$$\frac{2\pi}{T} = \sqrt{\frac{k}{m}}$$

$$\frac{T}{2\pi} = \sqrt{\frac{m}{k}}$$

$$T = 2\pi \sqrt{\frac{m}{k}}$$

$m \rightarrow$  inertial factor

$k \rightarrow$  spring factor



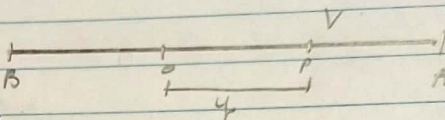
Equation for kinetic energy ( $E_K$ ).

$$E_K = \frac{1}{2} m v^2.$$

$$E_K = \frac{1}{2} m (\omega \sqrt{a^2 - y^2})^2.$$

$$E_K = \frac{1}{2} m \omega^2 (a^2 - y^2).$$

$$v = \omega \sqrt{a^2 - y^2}.$$



At mean position.

$$y = 0.$$

$$E_K = \frac{1}{2} m \omega^2 a^2 \rightarrow \text{max.}$$

At extreme position

$$y = \pm a.$$

$$E_K = \frac{1}{2} m \omega^2 (a^2 - a^2).$$

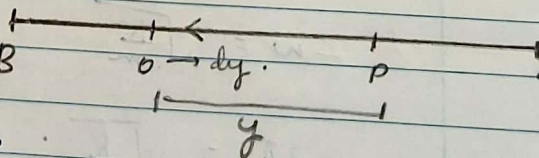
$$E_K = 0.$$

Equation for potential energy.

Minimum force to be applied.

$$F = ky.$$

Small workdone is moving the particle through small displacement  $dy$ .



$$F = -ky.$$



$$dw = F dy \cos 0^\circ$$

$$dw = ky dy$$

Total workdone.

$$dw = \int_0^y ky' dy$$

$$W = k \left( \frac{y^2}{2} \right)_0^y$$

$$W = \frac{1}{2} k (y^2)_0^y$$

$$W = \frac{1}{2} k (y^2 - 0^2)$$

$$W = \frac{1}{2} ky^2$$

This workdone is stored in the particle in the form of potential energy.

$$\therefore E_p = \frac{1}{2} ky^2$$

$$\text{But } k = m\omega^2$$

$$\therefore E_p = \frac{1}{2} m\omega^2 y^2$$

At mean position.

$$y = 0$$

$$E_p = 0$$

At extreme position

$$y = \pm a$$

$$E_p = \frac{1}{2} m\omega^2 a^2 - \text{maximum}$$



Equation for total energy

$$E_T = E_K + E_P$$

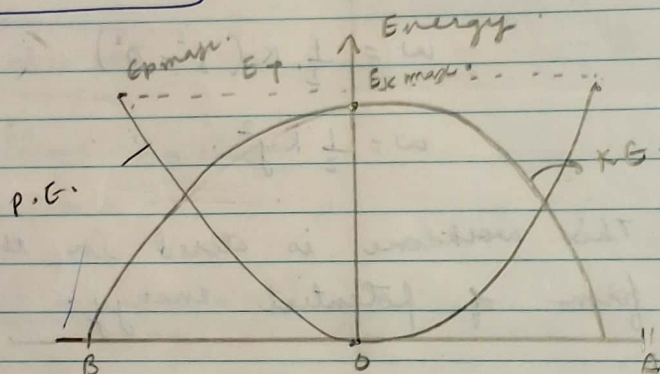
$$E_T = \frac{1}{2} m \omega^2 (a^2 - y^2) + \frac{1}{2} m \omega^2 y^2$$

$$E_T = \frac{1}{2} m \omega^2 a^2 - \frac{1}{2} m \omega^2 y^2 + \frac{1}{2} m \omega^2 y^2$$

clearly, of the particle  
Total energy remains constant.

$$E_T = \frac{1}{2} m \omega^2 a^2$$

$$E_T = \frac{1}{2} k a^2$$



Differential equation of simple harmonic motion.

we have.

$$F = -ky$$

$$F = mA$$

$$mA = -ky$$

$$mA + ky = 0$$

$$m \frac{dv}{dt} + ky = 0$$

$$m \frac{d}{dt} \left( \frac{dy}{dt} \right) + ky = 0$$

$$m \frac{d^2 y}{dt^2} + ky = 0$$



$$\frac{d^2 y}{dt^2} + \frac{k}{m} y = 0 \quad \left[ \begin{array}{l} k = m\omega^2 \\ \omega^2 = \frac{k}{m} \end{array} \right]$$

$$\frac{d^2 y}{dt^2} + \omega^2 y = 0$$

Its solution is  $y = a \sin \omega t$ .

### Simple Pendulum

we have,

$$F = -ky$$

also  $F = -mg \sin \theta$ .

$$-mg \sin \theta = -ky$$

$$mg \sin \theta = ky$$

we have  $\sin \theta = \frac{y}{l}$

$$\therefore mg \frac{y}{l} = ky$$

$$\frac{mg}{l} = k$$

we know that

$$T = 2\pi \sqrt{\frac{m}{k}}$$

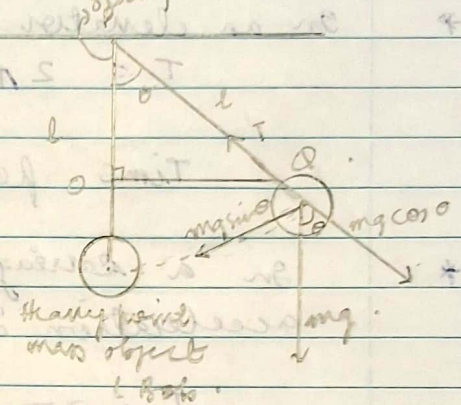
$$T = 2\pi \sqrt{\frac{mk l}{mg}}$$

$$T = 2\pi \sqrt{\frac{l}{g}}$$

At the center of earth

$$g = 0$$

$$\therefore T = 2\pi \sqrt{\frac{l}{0}} = \infty$$





- \* In an elevator moving vertically upwards with acceleration 'a'

$$T = 2\pi \sqrt{\frac{l}{g+a}}$$

Time period decreases.

- \* In an elevator moving downwards.

$$T = 2\pi \sqrt{\frac{l}{g-a}}$$

Time period increases.

- \* In a carriage moving horizontally with acceleration 'a'

$$T = 2\pi \sqrt{\frac{l}{g^2 + a^2}}$$

$T \rightarrow$  decreases.

### Second's Pendulum

It is a pendulum whose time period is 2 seconds.

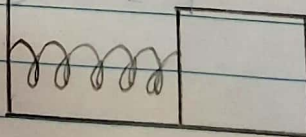
The length of the second's pendulum is 99.3 cm.

Oscillations of a loaded spring.

force on the load.

$$F = -kx$$

known as spring constant.





But  $F = mA$ .

$$\therefore F = m(-\omega^2 y)$$

$$F = -m\omega^2 y$$

$$m\omega^2 y = ky$$

$$m\omega^2 = k$$

$$\omega = \sqrt{\frac{k}{m}}$$

$$\frac{2\pi}{T} = \sqrt{\frac{k}{m}}$$

$$\boxed{\frac{T}{2\pi} = \sqrt{\frac{m}{k}}}$$

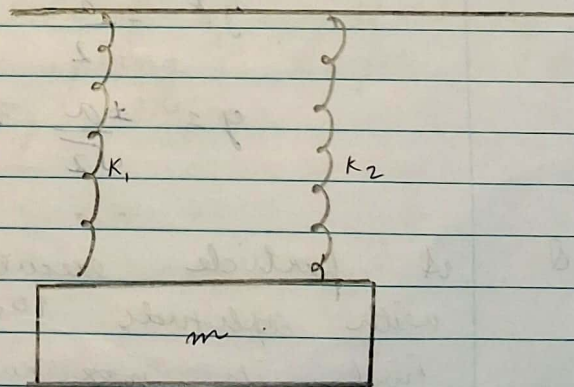
$$T = 2\pi \sqrt{\frac{m}{k}}$$

1. Springs in parallel.  
In parallel combination

$$K = k_1 + k_2$$

$$T = 2\pi \sqrt{\frac{m}{K}}$$

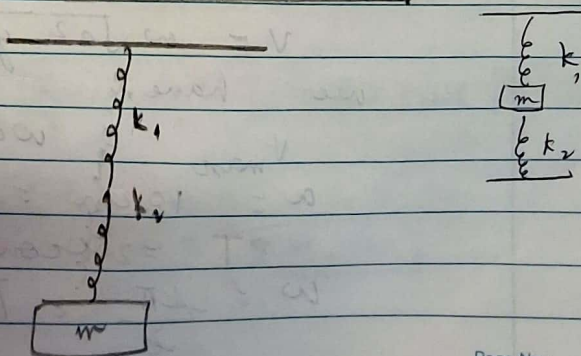
$$\Rightarrow T = 2\pi \sqrt{\frac{m}{k_1 + k_2}}$$



2. Springs in series.  
In series

$$\frac{1}{K} = \frac{1}{k_1} + \frac{1}{k_2}$$

$$\frac{1}{K} = \frac{k_1 + k_2}{k_1 k_2}$$





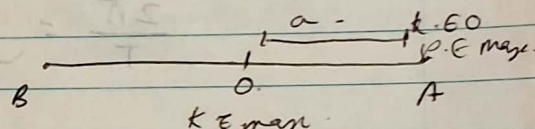
$$K = \frac{K_1 K_2}{K_1 + K_2}$$

$$\text{Now, } T = 2\pi \sqrt{\frac{m}{K}}$$

$$T = 2\pi \sqrt{\frac{m(K_1 + K_2)}{K_1 K_2}}$$

- Q. At what displacement from the mean position the kinetic energy of the particle is equal to its potential energy

$$E_K = E_P$$



$$\frac{1}{2} m \omega^2 (a^2 - y^2) = \frac{1}{2} m \omega^2 y^2$$

$$a^2 - y^2 = y^2$$

$$2y^2 = a^2$$

$$y^2 = \frac{a^2}{2} \quad \text{or} \quad y^2 = \frac{(\pm a)^2}{2}$$

$$y = \frac{\pm a}{\sqrt{2}}$$

- Q. A particle executes simple harmonic motion with amplitude 10 cm and time period 2 seconds. Find its maximum velocity.

$$V = \omega \sqrt{a^2 - y^2}$$

we have,

$$V_{\text{max}} = \omega a$$

$$a = 10 \text{ cm} = 0.1 \text{ m}$$

$$T = 2 \text{ seconds}$$

$$\omega = \frac{2\pi}{T} = \pi = 3.14 \text{ radian/sec}$$



$$\therefore v = 3.14 \times 0.1 = 0.314 \text{ m/s.}$$

- Q. A particle executes SHM having amplitude 20 cm. Find its maximum acceleration if its  $T = 0.2 \text{ sec}$ .

we know,

$$A_{\max} = -\omega^2 a.$$

$$a = 20 \text{ cm} = 0.2 \text{ m.}$$

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{0.2} = 10\pi \text{ rad/s.}$$

$$\begin{aligned} \text{now, } |A_{\max}| &= \omega^2 a \\ &= (10\pi)^2 \times 0.2 \\ &= 100 \times (3.14)^2 \times 0.2 \\ &= 20 \times 9.85 \\ &= 197.19 \approx 197.2 \text{ m/s}^2. \end{aligned}$$

- Q. A body of mass 10 kg is connected to two springs in parallel with each other. Find the time period of the loaded spring if spring constant ( $k$ ) of one spring is 2 N/m and that of another spring is 3 N/m.

$$T = 2\pi \sqrt{\frac{m}{k_1 + k_2}}$$

$$T = 2\pi \sqrt{\frac{m}{k}}$$

$$T = 2 \times 3.14 \sqrt{\frac{10}{2+3}} \quad \text{or} \quad T = 2 \times 3.14 \sqrt{\frac{10}{5}}$$

$$T = 2 \times 3.14 \times \sqrt{2}$$

$$T = 2 \times 3.14 \times 1.41 = \boxed{8.9, \text{ sec}}$$



- Q. A spring of spring constant  $k$  is cut into two equal halves. Find the spring constant of each part. Also what will be the spring constant of the combination if these two parts are connected in parallel with each other?

$$\frac{1}{k} = \frac{1}{k'} + \frac{1}{k'}$$

$$\frac{1}{k} = \frac{2}{k'}$$

$$k' = 2k$$

Now, they are connected in parallel.

$$k_T = k' + k'$$

$$k_T = 2k + 2k$$

$$k_T = 4k$$

Equation for displacement.

$$y = a \sin(\omega t \pm \phi)$$

Here  $\phi$  is initial phase and is known as the epoch.

Q. 1. In the  $y = 10 \sin(100\pi t + \pi/4)$  mm

Q. In the equation  $y = 10 \sin(100\pi t + \pi/4)$  mm then, find

- a) Amplitude, b)  $T$ , c) epoch, d) maximum velocity



comparing given eqn with  
 $y = a \sin(\omega t \pm \phi)$

$$y = a \sin(\omega t + \phi)$$

a)  $a = 10 \text{ cm.}$

b)  $\omega = 100\pi$

$$\frac{2\pi}{T} = 100\pi$$

$$T = \frac{2\pi}{100\pi} = 0.025$$

c)  $v_{\max} = \omega a$

$$v_{\max} = 100\pi \times 10 = 3140 \text{ m/s.}$$

d)  $|a_{\max}| = \omega^2 a = (100\pi)^2 \times 10$   
 $= 10^5 \times \pi^2$   
 $= 10^5 \times (3.14)^2$   
 $= 9.8 \times 10^5 \text{ m/s.}$

Q A particle is executing SHM with time period 4s. of the amplitude of oscillation is 5cm,

calculate:

1) displacement 2) velocity and 3) acceleration of the particle after 1 second it starts oscillating.

Sol. Here  $T = 4\text{s}$ ,  $a = 5\text{cm}$ ,  $t = 1\text{s}$ .

1 displacement  $y = a \sin \omega t = a \sin \frac{2\pi}{T} t$   
 $= 5 \sin \frac{2\pi}{4} = 5 \sin \frac{\pi}{2} = 5 \text{ cm.}$

2 velocity  $v = \omega \sqrt{a^2 - y^2} = \frac{2\pi}{T} \sqrt{25 - 25} = 0$ .

3 Acceleration  $|a| = \omega^2 y = \frac{4\pi^2}{T^2} y$   
 $= \frac{4\pi^2}{16} \times 5 = 5 \frac{\pi^2}{4} \text{ cm/s}^2$



Q2 A particle executes SHM of amplitude 30 cm and time period 4s. What is the minimum time required for the particle to move from mean position to a point 15 cm?

Let  $t$  be the time taken by the particle to move from mean position to a point 15 cm from it.  
we know,

$$y = a \sin \frac{2\pi}{T} t$$

Here  $y = 15 \text{ cm}$ ,  $a = 30 \text{ cm}$ ,  $T = 4 \text{ s}$ .

$$\therefore 15 = 30 \sin \frac{2\pi}{4} t$$

$$\text{or } \sin \frac{2\pi}{4} t = \frac{15}{30} = \frac{1}{2} = \sin \frac{\pi}{6}$$

$$\text{or } \frac{2\pi}{4} t = \frac{\pi}{6}$$

$$\text{or } t = \frac{1}{3} \text{ s}$$

Q3 The body of 1 kg mass is made to oscillate on a spring of force constant 9 N/m. Calculate the angular frequency of oscillation.

Here,  $m = 1 \text{ kg}$ ,  $K = 9 \text{ N/m}$ .

we know,

$$\omega = \left[ \frac{k}{m} \right]^{\frac{1}{2}} = \left[ \frac{9}{1} \right]^{\frac{1}{2}} = 3 \text{ rad}^{-1} \text{ s}^{-1}$$



Q4. A particle of mass  $10g$  is executing SHM with an amplitude of  $0.5m$  and periodic time of  $(\frac{\pi}{5})$  seconds. Calculate the maximum value of the force acting on the particle.

Here  $m = 10g = 10 \times 10^{-3} = 10^{-2} kg$ .

$a = 0.5m$   $T = \frac{\pi}{5} s$ .

Now,  $F = ma$ .

But  $a = \omega^2 a$  (max value of acc.)

$= \left(\frac{2\pi}{T}\right)^2 a$ .

$\therefore F = m \frac{4\pi^2}{T^2} a = \frac{10^{-2} \times 4\pi^2 \times 0.5 \times 25}{\pi^2}$   
 $= 0.5$ .

Q5. A particle executes SHM with an angular velocity of  $3.5 rad s^{-1}$  and maximum acceleration  $7.5 m/s^2$ . Find the amplitude of the oscillation.

Here  $\omega = 3.5 rad/s$ .

$|a_{max}| = 7.5 m/s^2$ .

$|a_{max}| = \omega^2 a$ .

$\therefore a = \frac{a_{max}}{\omega^2} = \frac{7.5}{(3.5)^2} = 0.61 m$ .

Q6. In what time after its motion begins will a particle oscillating according to the relation  $y = 7.5 \sin 0.5\pi t$ , move from the mean position to maximum displacement?

Here  $y = 7.5 \sin 0.5\pi t$ .



when a particle moves from mean position to maximum displacement,  $y = a = 7$ .

$$7 = 7 \sin 0.5 \pi t$$

$$\text{or } \sin 0.5 \pi t = \sin \frac{\pi}{2}$$

$$\text{or } 0.5 \pi t = \frac{\pi}{2}$$

$$\text{or } 0.5 t = \frac{1}{2}$$

$$t = 1.0 \text{ s.}$$

Q7. calculate the ratio of displacement to amplitude, when K.E of the body executing SHM is thrice the P.E.

$$\text{K.E of the body} = 3 \text{ times P.E}$$

$$\text{i.e.; } \frac{1}{2} m \omega^2 (a^2 - y^2) = 3 \times \frac{1}{2} m \omega^2 y^2$$

$$\text{or } a^2 - y^2 = 3 y^2 \text{ or } 4 y^2 = a^2$$

$$\text{or } 2y = a \text{ or } \frac{y}{a} = \frac{1}{2}$$

Q8. The time period of simple Pendulum is 2 seconds. what will be its time period if its length is increased to 4 times its original length?

we know,

$$T = 2\pi \sqrt{\frac{l}{g}}$$

$$\text{when } l' = 4l$$

$$\text{then } T' = 2\pi \sqrt{\frac{4l}{g}}$$

$$= 2 \times 2\pi \sqrt{\frac{l}{g}}$$

$$= 2T = 2 \times 2$$

$$= 4 \text{ s}$$



## Waves.

A wave is a disturbance which carries energy from one point to another point without the actual movement of matter from one point to another point.

→ Based on the requirement of medium, waves are of two types :-

### 1. Mechanical waves.

These are the waves which require material medium for their propagation.

e.g: sound waves.

### 2. Non-mechanical waves:

These are the waves which do not require any material medium for their propagation.

e.g: All electromagnetic waves (RMIVUXG).

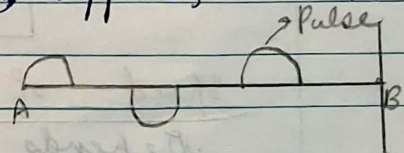
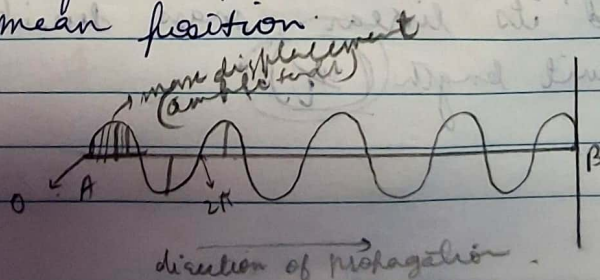
→ Depending upon the direction of vibration of particles, waves are of two types.

### 1. Transverse waves:

These are the waves in which the displacement of the particles (direction of vibration) is perpendicular to the direction of propagation of wave.

e.g: All electromagnetic waves, ripples, wave in a string.

1.1) Amplitude: It is the maximum displacement of the particle from the mean position.





1.2 One cycle: One crest plus one trough - (one wave).

1.3 Time period ( $T$ ): It is defined as the time taken by the wave to complete its one cycle.

1.4 Linear frequency ( $\nu$ ) =  $\frac{1}{T}$ . It represents no. of cycles per second.

1.5 Angular frequency =  $\frac{2\pi}{T}$ .

1.6 Wave length: ( $\lambda$ )  $\rightarrow$  It is defined as the length of one cycle of the wave. It is equal to the distance b/w two consecutive crests or the distance b/w two consecutive troughs.

1.7 Wave no: It is the reciprocal of wave length of the wave ( $\frac{1}{\lambda}$ ).

1.8 Angular wave no:  $k = \frac{2\pi}{\lambda}$ .

Speed of a wave:

$$v = \frac{\lambda}{T}$$

$$v = \lambda \times \frac{1}{T}$$

$$v = \lambda \times \nu$$

$$v = \lambda \nu$$

Speed of a transverse wave in a string  
Depends on the tension ( $T$ ) in the string and its linear mass density ( $m$ ) i.e. mass per unit length ( $\frac{m}{l}$ ).



$$v \propto T^a$$

$$v \propto m^b$$

$$v \propto T^a m^b$$

$$v = K T^a m^b \rightarrow \textcircled{1}$$

using principle of homogeneity of dimensions.

$$L^1 T^{-1} = (m^1 L^1 T^{-2})^a (m^1 L^{-1})^b$$

$$L^1 T^{-1} = m^a L^a T^{-2a} m^b L^{-b}$$

$$M^0 L^1 T^{-1} = M^{a+b} L^{a-b} T^{-2a}$$

comparing powers.

$$-2a = -1$$

$$a = \frac{1}{2}$$

$$\text{Also } a + b = 0$$

$$\frac{1}{2} + b = 0$$

$$\frac{1}{2} b = -\frac{1}{2}$$

From equation  $\textcircled{1}$

$$v = K T^{\frac{1}{2}} m^{-\frac{1}{2}}$$

$$v = K \frac{T^{\frac{1}{2}}}{m^{\frac{1}{2}}} = K \left( \frac{T}{m} \right)^{\frac{1}{2}}$$

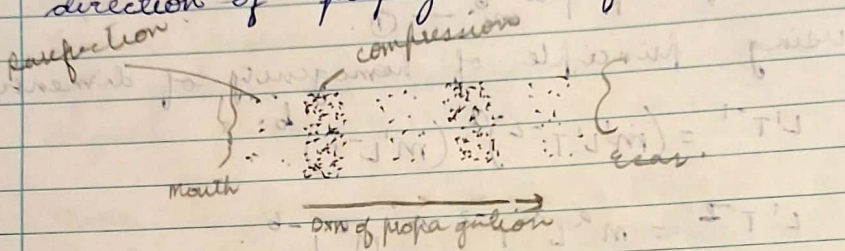
$$v = K \sqrt{\frac{T}{m}}$$

$$K = 1$$

$$\therefore v = \sqrt{\frac{T}{m}}$$



- 2 Longitudinal waves (sound waves): these are the waves in which the displacement of the particles (direction of vibration) is parallel to the direction of propagation of wave.



compression: Region in which density of particles is max. and therefore pressure is maximum.

rarefaction: Region in which density of particles is minimum and hence pressure is minimum.

- One cycle: one compression + one rarefaction.
- Time period: (T) time taken to complete one cycle.
- Linear frequency ( $\nu$ ):  $\frac{1}{T}$ .
- Angular frequency:  $\frac{2\pi}{T} = 2\pi\nu$ .
- wave no.  $\frac{1}{\lambda}$ .
- Angular wave no (k):  $\frac{2\pi}{\lambda}$ .
- wave length ( $\lambda$ ): Distance  $\lambda$  b/w two consecutive compressions or rarefactions.

Speed of a sound wave: The speed of a sound wave in a medium is given by.

$$v = \sqrt{\frac{E}{\rho}}, \text{ where } E \text{ is modulus of}$$



elasticity of the medium and  $\rho$  is its density. Although density of solids is high but value of  $E$  for solids is higher, therefore, speed of sound is maximum in solids.

$$v_{\text{solids}} > v_{\text{liq}} > v_{\text{gas}}$$

Speed of sound in a gas is given by:

$$v = \sqrt{\frac{\gamma P}{\rho}}$$

$$\frac{C_p}{C_v}$$

$$\gamma \text{ for air} = 1.4$$

$$v = \sqrt{\frac{\gamma P \bar{v}}{M}}$$

$$v = \sqrt{\frac{\gamma R T}{M}}$$

$$* \quad v \propto \sqrt{T}$$

As  $v \propto \sqrt{T}$ , therefore speed of sound is maximum in hot summer days as compared to the cold winter days.

Audible sound  $\rightarrow 20\text{Hz}$  to  $20\text{kHz}$

infrasonic sound  $\rightarrow < 20\text{Hz}$

ultrasonic sound  $\rightarrow > 20\text{kHz}$

At  $20^\circ\text{C}$  speed of sound in air is  $342\text{ m/s}$ .



Any object moving with a speed greater than speed of sound in air is said to be moving with **SUPERSONIC** speed.

NOTE:

□ In a wave, the particles of the medium execute simple harmonic oscillations about their mean position.

□ For a wave to propagate through a medium, the medium must possess two important properties

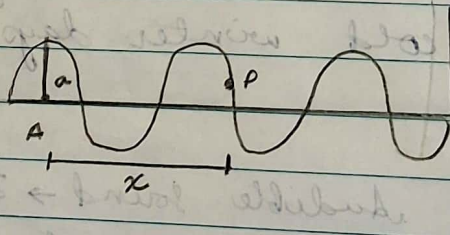
- 1) Elasticity
- 2) Inertia.

Equation of a plane progressive wave.

A progressive wave is a wave which carries energy from one point to another point.

At point A.

$$y = a \sin \omega t$$



At point P.

$$y = a \sin(\omega t - \Phi)$$

It means point P lags behind point A by phase of  $\Phi$

$$\text{Here } \Phi = kx$$



$$y = a \sin(\omega t - kx) \quad *$$

If wave propagates along -ve x axis  
 $y = a \sin(\omega t + kx)$ .

Other forms:

$$y = a \sin(\omega t - kx)$$

$$y = a \sin\left(\frac{2\pi t}{T} - \frac{2\pi x}{\lambda}\right) \quad \text{--- (1)}$$

$$y = a \sin\left(2\pi \nu t - \frac{2\pi x}{\lambda}\right) \quad \text{--- (2)}$$

$$y = a \sin\left[2\pi\left(\nu t - \frac{x}{\lambda}\right)\right] \quad \text{--- (3)}$$

\* A progressive wave is also known as a travelling wave.

Speed of a progressive / travelling wave.

$$v = \nu \lambda$$

$$v = \frac{\lambda}{T}$$

$$v = \frac{2\pi}{T} \times \frac{\lambda}{2\pi}$$

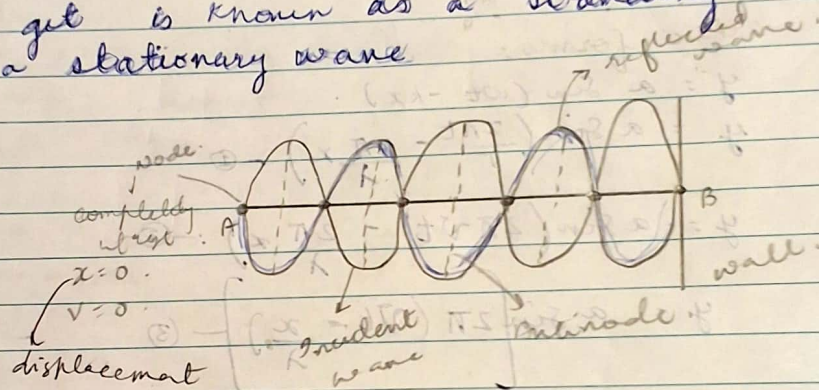
$$v = \frac{2\pi/T}{2\pi/\lambda} = \frac{\omega}{k}$$

$$v = \frac{\omega}{k}$$



V<sup>∞</sup> inf.

Standing waves or stationary waves, when two identical waves (same nature, same amplitude, same frequency and same wavelength) travelling in opposite directions super impose (to get combined); the resultant we get is known as a standing wave or a stationary wave.



There is no net transfer of energy in a standing wave.

**Nodes:** These are the points in a standing wave where the ~~displacement~~ displacement of the particles is zero i.e; velocity of the particles is zero.

**Antinodes:** These are the points where the displacement of the particles is maximum. i.e; velocity is maximum.

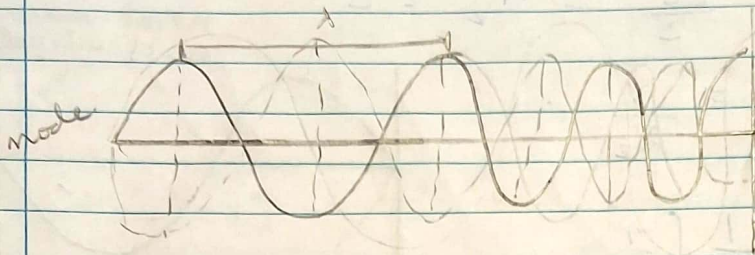
$$\frac{\omega}{k} = \frac{T/\Delta t}{\Delta x/\Delta t} = V$$

✓

$$\frac{\omega}{k} = V$$



Standing waves in a string fixed at both the ends



Distance between two consecutive nodes =  $\frac{\lambda}{2}$   
 Distance between two consecutive antinodes =  $\frac{\lambda}{2}$

Distance between a node and an antinode =  $\frac{\lambda}{4}$

Normal modes of vibration -

$$\lambda_n = \frac{2L}{n}, \quad n = 1, 2, 3, \dots$$

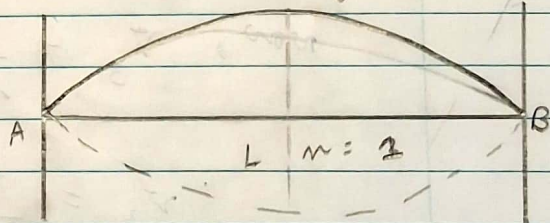
1st mode  $n = 1$

no. of nodes = 2

no. of antinodes = 1

$$\lambda_1 = \frac{2L}{1}$$

$$L = \frac{\lambda_1}{2}$$



$$\text{now, } v_1 = \frac{v}{\lambda}$$

$$v_1 = \frac{1}{2L} \sqrt{\frac{T}{m}}$$

This frequency is known as fundamental frequency or 1st harmonic.

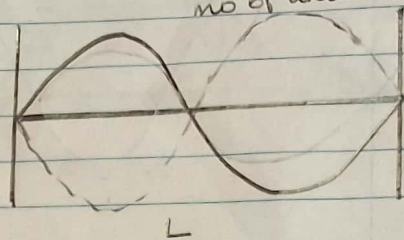


2nd mode  $n = 2$

$$\lambda_2 = \frac{2L}{2} \therefore \lambda_2 = L$$

now,  $v_2 = \frac{v}{\lambda_2}$

$$v_2 = \frac{1}{L} \sqrt{\frac{T}{m}}$$



This frequency is known as 2nd harmonic, or 1 overtone

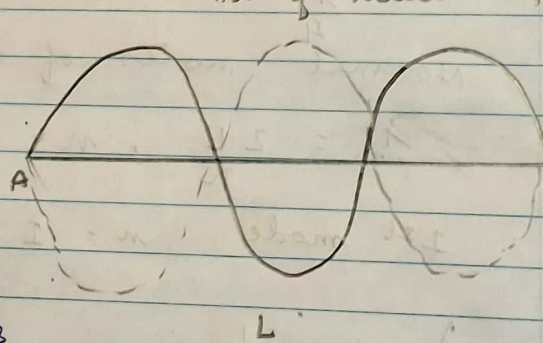
3rd mode  $n = 3$

$$\lambda_3 = \frac{2L}{3}$$

$$L = \frac{3\lambda_3}{2}$$

now,  $v_3 = \frac{v}{\lambda_3}$

$$v_3 = \frac{3}{2L} \sqrt{\frac{T}{m}}$$



This frequency is known as third harmonic or 2nd overtone.

In general  $n = n$   
For  $n^{\text{th}}$  mode

$$\lambda_n = \frac{2L}{n}$$

$$L = \frac{n\lambda_n}{2}$$



no. of antinodes =  $n$

no of nodes =  $n + 1$

$$v_n = \frac{v}{\lambda_n} = \frac{1}{\frac{2L}{n}} \sqrt{\frac{T}{m}}$$

$$v_n = n v_1$$

This frequency is known as  $n$ th harmonic  
or  $(n-1)$ th overtone.

Q. The fundamental frequency in a string fixed at both the ends is  $400 \text{ Hz}$ . what is the frequency of 7th overtone?

For 7th overtone =  $n = 8$ .

$$v_8 = 8 v_1$$

$$v_8 = 16 \text{ Hz}$$



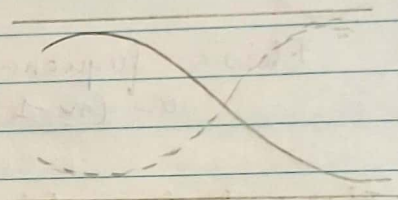
standing waves in an organ pipe open at both the ends.

$$\lambda_n = \frac{2L}{n}$$

First mode of vibration  
 $n=1$ .

$$\lambda_1 = \frac{2L}{1}$$

$$L = \frac{\lambda}{2}$$



Also  $v = \nu \lambda$

$$\nu_1 = \frac{v}{\lambda_1}$$

$n=1$

no of nodes = 1

no of antinodes = 2

$$\nu_1 = \frac{v}{2L}$$

This frequency is known as fundamental frequency or first harmonic.

2nd mode of vibration,  
 $n=2$ .

$$\lambda_2 = \frac{2L}{2}$$

$$\lambda_2 = L$$

frequency

$$\nu = \nu_2 \lambda_2$$

$$\nu = \frac{v}{\lambda_2}$$

$$\nu = \frac{v}{L}$$



This frequency is known as 2nd harmonic or 1st overtone.

3rd mode of vibration

$$n = 3$$

$$\lambda_3 = \frac{2L}{3}$$

$$L = \frac{3\lambda_3}{2}$$

frequency.

$$v = v_3 \lambda_3$$

$$v = \frac{v}{\lambda_3}$$

$$v = \frac{v}{2L} \times 3$$

$$= \frac{3v}{2L}$$

This frequency is known as third harmonic or second overtone.

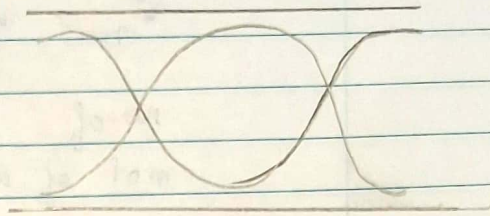
$v_{n\text{th}}$  =  $n^{\text{th}}$  mode of vibration.  
 $n = n$

$$\lambda_n = \frac{2L}{n}$$

$$L = \frac{n\lambda_n}{2}$$

$$\text{now, } v = v_n \lambda_n$$

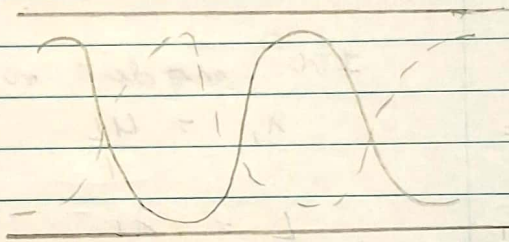
$$v_n = \frac{v}{\lambda_n}$$



$$n = 2$$

no of nodes = 2

no of antinodes = 3



$$n = 3$$

no of nodes = 3

no of antinodes = 4



$$v_n = \frac{nV}{2L}$$

$$v_n = n v_1$$

no of nodes =  $n$   
 no of antinodes =  $n+1$ .

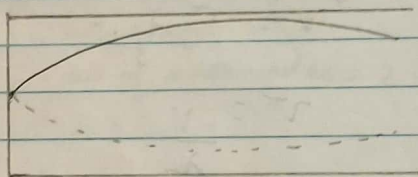
standing waves in an organ pipe closed at one end.

$$\lambda_n = \frac{4L}{2n-1}$$

1st mode:  $n=1$

$$\lambda_1 = \frac{4L}{1}$$

$$L = \frac{\lambda_1}{4}$$



$$\text{now, } v = v_1 \lambda_1$$

$$v_1 = \frac{v}{\lambda_1}$$

$$v_1 = \frac{v}{4L}$$

This frequency is known as fundamental frequency or 1st harmonic.

2nd mode  $\Rightarrow n=2$

$$\lambda_2 = \frac{4L}{2 \cdot 2 - 1}$$

$$\lambda_2 = \frac{4L}{4-1}$$

$$\lambda_2 = \frac{4L}{3}$$



Now  $v = v_2 \lambda_2$

$$v_2 = \frac{v}{\lambda_2}$$

$$v_2 = \frac{v}{2L}$$

Now,  $v = v_2 \lambda_2$

$$v_2 = \frac{v}{\lambda_2}$$

$$v_2 = \frac{3v}{4L}$$

$$v_2 = 3v_1$$

This frequency is known as 3rd harmonic or 1st overtone.

3rd mode:

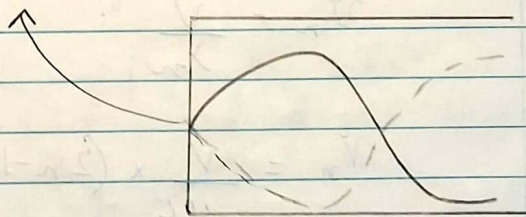
$$n = 3$$

$$\lambda_3 = \frac{4L}{2 \cdot 3 - 1}$$

$$\lambda_3 = \frac{4L}{6-1}$$

$$\lambda_3 = \frac{4L}{5}$$

$$L = \frac{5\lambda_3}{4}$$



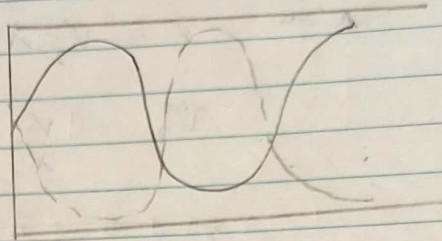
no of nodes = 2.  
no of antinodes = 2.



$$v = v_3 \lambda_3$$

$$v_3 = \frac{v}{\lambda_3}$$

$$v_3 = \frac{5v}{4L}$$



no. of nodes = 3  
no. of antinodes = 3.

$v_3 = 5v_1$ .  $\rightarrow$  this frequency is known as 5th harmonic or second overtone.

nth mode -

$$n = n$$

$$\lambda_n = \frac{4L}{2n-1}$$

$$v = \lambda_n v_n$$

$$v_n = \frac{v}{\lambda_n}$$

$$v_n = \frac{v}{4L} \times (2n-1)$$

$$v_n = (2n-1) v_1$$



### Beats.

It is the phenomenon of <sup>alternate</sup> rise and fall in the intensity of ~~sound~~ sound when two nearly same amplitude and nearly equal frequencies going along same direction superimpose with each other.

of at  $t=0$ , one beat is said to be formed when the intensity becomes maximum again with one minimum in between.

The time taken to form one beat is known as beat period and the reciprocal of beat period is known as beat frequency.

VV9mf

If  $\nu_1$  and  $\nu_2$  are the frequencies of two sound waves, then beat frequency is given by:

$$f = \nu_1 - \nu_2$$

$$T = \frac{1}{\nu_1 - \nu_2}$$

and beat period is given by.

Sound wave 1.

$$y_1 = a \sin 2\pi \nu_1 t$$

Sound wave 2

$$y_2 = b \sin 2\pi \nu_2 t$$



$$\left. \begin{aligned} \sin A + \sin B &= 2 \sin \left( \frac{A+B}{2} \right) \cos \left( \frac{A-B}{2} \right) \end{aligned} \right\}$$

$$y = a \sin 2\pi \nu_1 t + a \sin 2\pi \nu_2 t.$$

$$y = a [\sin 2\pi \nu_1 t + \sin 2\pi \nu_2 t]$$

$$y = a \left[ 2 \sin \left( \frac{2\pi \nu_1 t + 2\pi \nu_2 t}{2} \right) \cos \left( \frac{2\pi \nu_1 t - 2\pi \nu_2 t}{2} \right) \right]$$

$$y = 2a [\sin \pi (\nu_1 + \nu_2) t \cos \pi (\nu_1 - \nu_2) t]$$

$$y = 2a \cos \pi (\nu_1 - \nu_2) t \sin \pi (\nu_1 + \nu_2) t$$

$$y = A \sin \pi (\nu_1 + \nu_2) t$$

$$\text{where } A = 2a \cos \pi (\nu_1 - \nu_2) t$$

represents amplitude of resultant wave.

→ Greater the amplitude, greater is the intensity.

For. max intensity

$$A = \text{max}$$

$$\cos (\text{even multiple of } \pi) = 1$$

$$\cos (\text{odd multiple of } \pi) = -1$$

$$2a \cos \pi (\nu_1 - \nu_2) t = \text{max}$$

$$\Rightarrow \cos \pi (\nu_1 - \nu_2) t = \pm 1$$

$$\cos \pi (\nu_1 - \nu_2) t = \cos n\pi$$

where  $n = 0, 1, 2, 3, \dots$



$$\pi(\nu_1 - \nu_2)t = n\pi$$

$$t = \frac{n}{\nu_1 - \nu_2}$$

Intensity is max at .

$$t = 0, \frac{1}{\nu_1 - \nu_2}, \frac{2}{\nu_1 - \nu_2}, \frac{3}{\nu_1 - \nu_2}, \frac{4}{\nu_1 - \nu_2} \dots$$

Beat period

$$T = \frac{1}{\nu_1 - \nu_2} \dots$$

$$T = \frac{1}{\nu_1 - \nu_2}$$

$$f = \frac{1}{T} = \nu_1 - \nu_2$$

How to find the percentage increase in the time period of a simple pendulum due to percentage increase in its length.

Formula to be used :-  $\Delta T\% = \left[ \sqrt{1 + \frac{\Delta L\%}{100}} - 1 \right] \times 100\%$

where  $\Delta L\%$  represents percentage increase in length and  $\Delta T\%$  represents percentage increase in time period.



Q. If the length of the simple pendulum is increased by 10% what will be the percentage increase in its time period.

$$\Delta L\% = 10.$$

we have,

$$\Delta T\% = \frac{\sqrt{1 + \Delta L\%}}{100} - 1 \times 100\%$$

$$= \frac{\sqrt{1 + 10}}{100} - 1 \times 100\%$$

$$= \sqrt{1.1} - 1 \times 100\%$$

$$= (1.0488 - 1) \times 100\%$$

$$= 0.0488 \times 100\%$$

$$= 4.88\%.$$

of Relation between K.E 'E' and Linear momentum 'p'.

$$K.E = \frac{1}{2} m v^2.$$

$$E = \frac{1}{2} \frac{m^2 v^2}{m}$$

$$E = \frac{1}{2} \frac{(mv)^2}{m}$$



$$E = \frac{1}{2} p^2$$

$$E = \frac{p^2}{2m}$$

Q. A light body and a heavy body have same linear momentum which one has greater kinetic energy.  $[E = \frac{p^2}{2m}]$ .

From the above formula, as  $E \propto \frac{1}{m}$ .  
 $\therefore$  light body has greater K.E.

$$p^2 = 2mE$$

$$p = \sqrt{2mE}$$

Q. A light body and a heavy body have same K.E. which one has greater linear momentum.

$$\text{As } p \propto \sqrt{m}$$

Clearly, heavy body has greater linear momentum.

concept: If  $x$  is the fractional increase in linear momentum of a body, then the percentage increase in its K.E. is given by

$$\Delta E\% = [(1+x)^2 - 1] \times 100\%$$

Q. If the linear momentum of a particle is increased by 50% what is the percentage increase in its K.E.?

$$\left[ \text{fractional} = \frac{\% \text{age}}{100} \right]$$

%age increase in  $p = 50$



$$\text{fractional inc in } p = x = \frac{50}{100} = 0.5.$$

$$\therefore \Delta B\% = \left[ (1 + 0.5)^2 - 1 \right] \times 100\%.$$

$$= \left[ (1.5)^2 - 1 \right] \times 100\%.$$

$$= (2.25 - 1) \times 100\%.$$

$$= 1.25 \times 100\%$$

$$= 125\%.$$

### TUNING FORK PROBLEM.

If  $\nu_1$  is the known frequency of tuning fork,  $\nu_2$  is the unknown frequency of tuning fork and 'f' is the beat frequency, then

$$\nu_2 = \nu_1 \pm f.$$

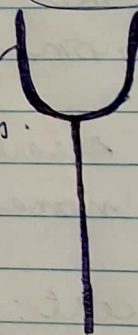
$$5 = 250 - x$$

$$5 - 250 = x$$

$$-245 = x$$

$$5 = x - 250$$

$$x = 255$$



on loading the tuning fork with wax (unknown), if beat frequency decreases then,  $\nu_2 = \nu_1 + f$ .

on loading the tuning fork with wax (unknown), if the beat frequency increases then,  $\nu_2 = \nu_1 - f$ .



- Q. A tuning fork produces 4 beats per second with a tuning fork of frequency  $288 \text{ Hz}$ . A little wax is placed on the unknown tuning fork and it then produces 2 beats per second. What is the frequency of unknown tuning fork.

As beat freq decreases

$$\nu_2 = \nu_1 + f$$

$$\nu_2 = 288 + 4$$

$$\nu_2 = 292 \text{ Hz}$$